

The Research of Power Allocation in Diesel-electric Hybrid Propulsion System

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Abstract—In the traditional diesel engine propulsion mode, the ships face up to a serial of issues, such as the inefficient performance of marine main engine under low load, the high fuel consumption rate and inferior combustion quality, which hinder the income increase and the pollution decrease. Diesel-electric hybrid propulsion ships can effectively reduce fuel consumption and emissions through reasonable power distribution. Moreover, it also has excellent handling performance and low noise. Taking a certain Deep Water Anchor Handling Tug Supply Vessel as the object, the multi-objective genetic algorithm (NSGA-II) was used to optimize the allocation problem of the power system. The optimization calculation and simulation verification were carried out by using MATLAB.

Keywords—Diesel-electric Hybrid ship; Power Allocation; Multi-objective Optimization; Non-dominated Sorting Genetic Algorithm Introduction

I. INTRODUCTION

Hybrid Propulsion Vessel(HPV) means that a single ship contains two or more types of power, such as sail-assisted propulsion^[1], water jet propulsion^[2], diesel-electric hybrid propulsion^[3] and so on. The diesel-electric hybrid system is one of the main forms of the hybrid propulsion system, which uses the rudder-paddle as the main propulsion mode. The electric propulsion mode is introduced in the traditional diesel propulsion system to improve the handling performance of the ship. The advantage is that Clean Energy (solar energy, wind energy), or fuel cells and batteries can be power sources^[4]. Additionally, the advantages of high power and high output torque of the diesel propulsion unit can be adopted. The control issues of hybrid propulsion system mainly focuses on how to improve the power configuration scheme under various working modes and achieves the goal of reducing fuel cost, decreasing emissions and improving ship economy under the premise of meeting power demand.

Hybrid ships and vehicles are very similar in terms of power form. They essentially use a variety of power sources (diesel, gasoline, batteries, etc.) to drive the operation of a ship or car^[5-8]. At present, domestic and foreign scholars have made considerable achievements in the research of hybrid electric vehicles, which has reference significance for the study of hybrid ships. Lan et al. was studied with a photovoltaic cell (PV), diesel engine and battery hybrid ship. The article separately carried on the modeling of the three parts and combined with the method of multi-objective particle swarm optimization (MOPSO) and non dominated sorting genetic algorithm (NSGA-II), optimized the shipping

economy and CO₂ emissions at the same time, it is concluded that the optimal size of PV and battery and capacity^[9]. In the ship-related research, Zehedi et al. considered the change of electric load of the ship under various working conditions. They proposed a control strategy to minimize the fuel consumption rate and formulated the generator set according to the average required power and power variation characteristics of the ship. The working principle of the strategy is to keep the work in an optimal working state. The optimized DC power system with energy storage device significantly reduces fuel consumption compared with the AC system^[10]. Mehdi et al. used a particle swarm optimization algorithm with constraints to design the optimal output power combination of the prime mover, generator set and other power sources^[11].

This paper takes the Deep Water Anchor Handling Tug Supply Vessel (DW-AHT) hybrid system as the research object, and studies the optimal distribution scheme of the power system under different working conditions to meet the demands of saving energy, high efficiency and low emissions. Compared to the performance requirements of ordinary passenger ships or special-purpose ships, the deep-water three-purpose work ship should take into account the working conditions such as navigation, positioning and towing^[11]. In this paper, energy saving and emission will be the target parameters. Using A Fast and Elitist Multiobjective Genetic Algorithm (NSGA-II) and mixed integer programming to calculate all Pareto optimal solutions and obtain the Pareto frontier. Then, using the Fuzzy decision algorithm method to get the optimal power allocation scheme for each working mode. This paper attempts to obtain the relevant theoretical basis for optimizing the actual ship power distribution, and then achieve the goal of improving the economy of the ship and reducing the environmental impact.

II. HYBRID PROPULSION SYSTEM MODEL

The Deep Water Anchor Handling Tug Supply Vessel (DW-AHT) propulsion system consists of two rudder propeller propulsion systems directly driven by two independently operated main diesel engines, two electric-driven all-direction propellers (AP) and three bow thrusters, as shown in Figure 1. The power system consists of two main engines SG1, SG2 and four diesel engine (DE) generator sets G3~G6. The main switch is used to supply the high voltage bus bars Bus1 and Bus2. The two full swing propellers M6 and M7 are connected to Bus1 and Bus2 respectively. Partition power supply is realized, and the busbars at both ends are separated by a zone switch. The M4 and M5 two bow thrusters (BT) and the M3 side thruster can provide lateral thrust for the ship. The main diesel engine (MDE) is mainly

used to drive the #1 and #2 main propellers, and can also drive the shaft generators G1 and G2 to supply the busbars. The propulsion motors M1 and M2 drive main propellers (MP) through the #1 and #2 main gearboxes (MG) respectively. The position of each thruster is represented by the coordinate value of the ship's core as the origin and the starboard direction is the positive direction of the Y-axis, as shown in Fig. 2.

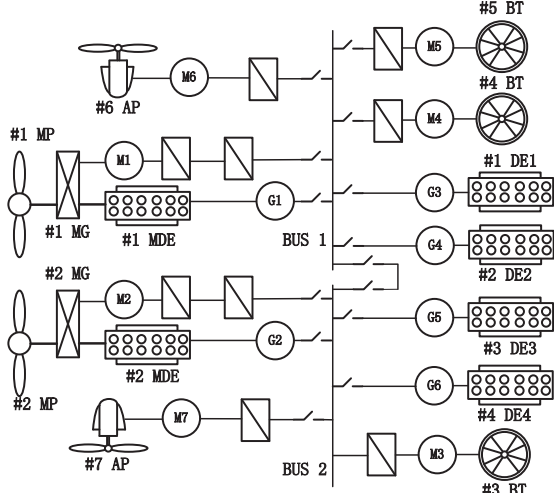


Fig.1 Deep water three-purpose work boat power system structure diagram

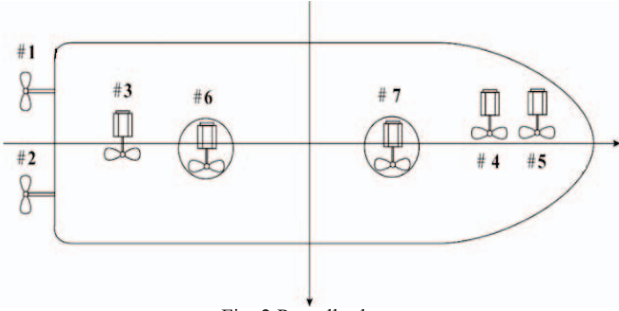


Fig. 2 Propeller layout

The power system can work in PTO (Power Take Off) mode, PTH (Power Take me Home) mode and BOOST mode respectively, corresponding to the different propulsion conditions of the ship. The PTO mode is the shaft generator mode. In this mode, the main diesel engine drives the main propeller to work, while the surplus power drives the shaft generators SG1 and SG2 to generate electricity for the ship's power grid. PTH mode and BOOST mode are usually called PTI (Power Take In) mode. PTH mode is suitable for low speed sailing conditions. At this time, the required propulsive force of the ship is small, the main diesel engine does not work, and the shaft motor is used as the motor at this time to drive the main thruster. The Boost mode is that the propulsion power of the main propeller can be further improved by the gear box when the thrust generated by the main diesel engine directly driving the propeller cannot meet the requirements.

TABLE I THREE WORKING STATES OF THE DW-AHT

Operating mode	Working state of shaft motor	Main diesel engine working condition
PTO	working as a generator	On
PTH	working as a motor	Off
BOOST	working as a motor	Off

III. OBJECTIVE FUNCTION

A. Minimize fuel consumption costs of hybrid propulsion systems

Under the normal operating conditions, the fuel consumption cost of the hybrid propulsion system includes the main fuel consumption cost and the generator fuel consumption cost. The prime movers of the main engine and the generator set belong to the diesel engine. The hourly fuel consumption curve of the diesel engine can be approximated by a quadratic polynomial function about the load:

$$g(P_i) = c_i + b_i P_i + a_i P_i^2 \quad (1)$$

$g(P_i)$ is the hourly fuel consumption of the i -th diesel engine, unit for g/h; a_i , b_i and c_i are the polynomial coefficients of the i -th diesel engine; P_i is the load power of the i -th diesel engine, unit for kW.

The fuel consumption rate refers to the fuel consumption per hour of power per unit of diesel engine, which can be expressed by (1):

$$SFOC(P_i) = P_i^{-1} g(P_i) \quad (2)$$

According to the relevant technical manual, the main engine generally uses heavy oil, and the generator set uses light diesel. Considering the difference between the prices of the two fuels, the fuel consumption cost of the diesel engine can be expressed as (3):

$$Cost_i' = 10^{-6} \cdot g(P_i) \cdot Cost_i \quad (3)$$

The total fuel consumption of a hybrid propulsion system can be expressed as:

$$\min f_1 = F(P_i) = 10^{-6} \sum_{i=1}^N s_i \cdot g(P_i) \cdot Cost_i \quad (4)$$

Where $F(P_i)$ is the fuel consumption cost of the hybrid propulsion system, the unit is \$/h; N is the number of diesel engines; s_i is the operating state of the i -th diesel engine, $s_i = 1$ when running, otherwise is 0; $Cost_i$ is the fuel price consumed for the i -th diesel engine, in units of \$/t.

B. Minimize emissions of pollutant gases

Controlling the emission of polluting gases is significant for ships to reduce air pollution and meet legal and regulatory requirements. This paper selects the emission of NOx from ships be the target. The hourly NOx emissions curve for a diesel engine can be approximated by a quadratic polynomial function with respect to the load:

$$e(P_i) = \gamma_i + \beta_i P_i + \alpha_i P_i^2 \quad (5)$$

Where $e(P_i)$ is the hourly emissions of the i diesel engine in g/h; α_i , β_i and γ_i are the emission polynomial coefficients of the i -th diesel engine. The total hourly NOx emissions $E(P_i)$ (t/h) of the ship can be expressed by the following formula:

$$\min f_2 = E(P_i) = 10^{-6} \sum_{i=1}^N s_i \cdot e(P_i) \quad (6)$$

IV. RESTRICTIONS

A. Diesel engine force constraint

Under normal operating conditions, different operating power limits should be set as (7) for different diesel engines, to prevent overload or operation at excessively low loads and the damage.

$$P_i^{\min} \leq P_i \leq P_i^{\max} \quad (7)$$

Where $P_i^{\max}$ and $P_i^{\min}$ respectively represent the upper and lower limits of the output power of the i -th diesel engine, and the unit is kW.

B. Ship power balance constraint

If the power loss in the system is not taken into account, the power delivered by the marine hybrid system should be balanced with the power required by the ship. The power required by the ship includes the power used for propulsion and the power supplied to the electrical load shown in (8).

$$P_{prop} + P_{elec} = P_{total} = \sum_{i=1}^N P_i \quad (8)$$

P_{prop} , P_{elec} and P_{total} are propulsion power, power generation, and total power required by the ship, respectively, the unit is kW.

C. Propulsion motor power constraint

The power output by the gensets includes power for propulsion and power for power generation. The power used by the generator set for propulsion shall not exceed the maximum output power that the propulsion motor can output. In addition, in the limit case, assuming that the power used for power generation is entirely borne by the generator set, there are:

$$\sum_{i=1}^{N_G} s_i \cdot P_i \leq N_{PM} P_{PM}^{\max} + P_{elec} \quad (9)$$

Where N_G is the number of generators; N_{PM} is the number of propulsion motors; $P_{PM}^{\max}$ is the maximum output power of a single propulsion motor, in kW.

V. OPTIMIZATION

A. NSGA-II

In multi-objective optimization, there might exist be mutual constraints between targets. Improving one target can bring positive influences on other targets. Therefore, the solution to the problem is often represented by the Pareto frontier. The solution on the frontier of Pareto is also called non-dominated solution. According to the working characteristics of the hybrid ship's power system, this paper uses NSGA-II to solve the Pareto frontier. NSGA-II is a Multi-objective Evolutionary Algorithm (MOEA) improved on the basis of NSGA. It adopts fast non-dominated sorting and elite strategy to speed up the convergence process in the search process and the rank. The same individuals choose to retain according to the crowding mechanism, which can effectively protect the diversity of the optimal solution.

The complete non-dominated sorting multi-objective genetic algorithm (NSGA-II) steps are as follows:

Step 1: Randomly generate the initial population P_0 , the population size is N , and calculate the fitness of the individual within the population, and set the evolution algebra to $t = 0$.

Step 2: Performing non-dominated sorting on the initial population P_0 . The fitness of each population is its non-dominated level. By sorting, each individual gets a RANK value and calculates the crowding distance.

Step 3: Selecting elite individuals through the competition method, and generate a parental population Q_0 whose population size is also N through crossover and mutation operations, and calculate the fitness as a non-dominated level again.

Step 4: Combining the progeny population Q_t and the parent population P_t to obtain a new population R_t with a population size of $2N$.

Step 5: Using R_t to compare the crowded comparison operators, and then select the best-ordered individuals to copy to the new population P_{t+1} until the new population size is N . At this time, it is a cycle, and the cycle algebra t add 1.

Step 6: Determine whether the maximum number of iterations $t_{\max}$ is reached. If not, repeat Step3-Step5 until the condition is met and the algorithm stops.

B. Fuzzy decision

After the multi-objective genetic algorithm is calculated, the resulting Pareto frontier is a set of solutions. In theory, this set of solutions can be considered as the optimal solution under the current conditions, called the optimal solution set. The method of fuzzy decision-making can be obtained from the set of optimal solutions, and the degree of membership of each solution is calculated by (10) by fuzzifying the optimal solution of the set.

$$\mu_i = \begin{cases} 1 & f_i < f_{i,\min} \\ \frac{f_{i,\max} - f_i}{f_{i,\max} - f_{i,\min}} & f_{i,\min} \leq f_i \leq f_{i,\max} \\ 0 & f_i > f_{i,\max} \end{cases} \quad (10)$$

μ_i is the objective function after fuzzification, f_i is the i -th solution of the optimal solution, $f_{i,\min}$ is the minimum value of the optimal solution, and $f_{i,\max}$ is the maximum value of the optimal solution.

After the fuzzy solution is fuzzified, all the optimal solution sets under a certain working condition constitute a fuzzy evaluation matrix. In order to facilitate the calculation and consider the authenticity of the simulation, for the four working conditions proposed, the fuzzy evaluation matrix is selected as shown in the table II:

TABLE II THE FUZZY EVALUATION MATRIX

Working Condition	HB	FS	DP	T
Fuzzy evaluation matrix	[0.1 0.9]	[0.8 0.2]	[0.5 0.5]	[0.5 0.5]

VI. OPTIMIZATION PROCESS AND SIMULATION RESULTS

A. Calculation process

The DW-AHT has six diesel engines, there are two main engines and three generators. The specific parameters are shown in Table III. In order to ensure the authenticity of the simulation work, the workload range of the ship's main engine and generator are determined to be between 30% and 100% in the process of power distribution. In the power distribution, if two main engines or two and more generators are running at the same time, the same loads should be ensure according to the working characteristics of the marine diesel engine,

TABLE III MAIN PARAMETERS OF THE MAIN ENGINE AND GENERATOR

Types	Number	Rated power /kW	Rated speed /rpm	Bore diameter /mm	Stroke /mm
Main Engine	2	8800	750	320	400
Generator	4	2500	1500	170	215

When the ship is sailing at a constant speed, the propeller thrust is balanced with the resistance of the ship.

The power after propeller can be obtained by calculating the resistance during navigation. Then by considering the power transfer efficiency of the propeller, the required propulsion power can be obtained^[12]. The total power required by a ship during navigation is the sum of the propulsion power and the power required by other loads on the ship. This paper considers the power distribution problem of the DW-AHT under the four conditions of towing, dynamic positioning, full-speed navigation and harbor in a certain travel mission. The ship speed map is shown in Fig.3.

Among them, the required power under the four working conditions is calculated as shown in the table IV where the calculation of the electric power is assumed to be a fixed value and only affected by the change of the working condition. The calculation method is based on the inspection of the ship and the power load is calculated.

TABLE IV THE REQUIRED POWER UNDER THE FOUR WORKING CONDITIONS

Mode	Propulsion power /kW	Electric power /kW	Totle power/kW
Tow (T)	10630	4000	14630
Dynamic positioning (DP)	0	5000	5000
Full speed (FS)	16708	2000	18708
Haber (HB)	0	700	700

B. Operating mode

Assuming that the ship's working condition and speed at 12h on a certain day is: 2 hours after the port is Harbor (HB), it will sail at full speed (FS) for 4 hours at an average speed of 18 kn to a certain place at sea, and then perform dynamic positioning (DP) 3 hours of operation, and finally with a towing force of 100 tons under the tow (T) condition sailing at an average speed of 8 kn for 3 hours. The specific speed change is shown in Fig. 3.

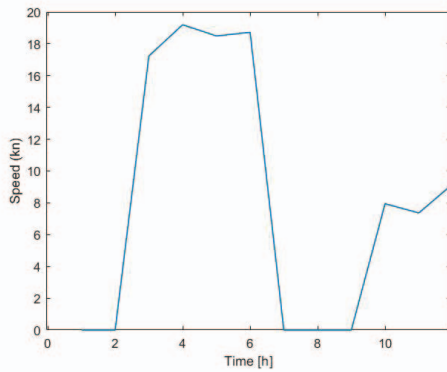


Fig.3 The speed map

C. Simulation results

Fig.4 shows the final Pareto frontier calculated using the power allocation algorithm for four different operating conditions. Each smoother continuous curve in full-speed navigation and towing conditions represents an effective solution. There are many Pareto optimal solutions in each effective scheme. The Pareto frontier of port-based mooring and dynamic positioning navigation conditions Medium is not a curve. However, one or several scattered points, each of which represents an effective solution. By using the fuzzy decision algorithm proposed in Chapter V, a solution that

more accurate meets the requirements can be selected among many Pareto optimal solutions. The final results are presented in Table V.

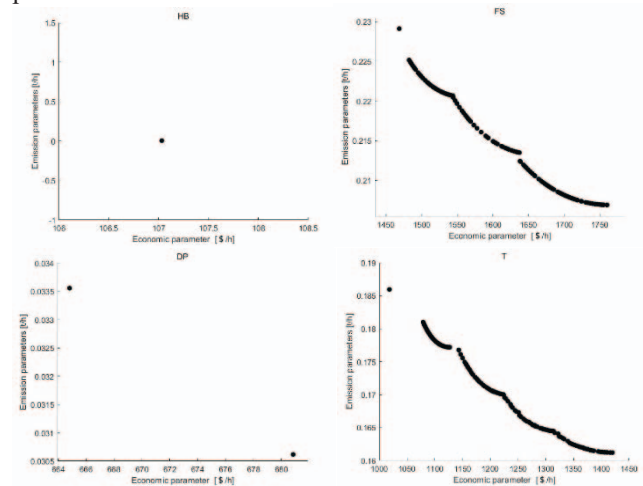


Fig.4 Pareto front for each working condition

TABLE.V OPTIMAL SOLUTIONS OBTAINED AFTER FUZZY DECISION

Objective function	HB	FS	T	DP
$f_1 / (\text{US } \$/\text{h})$	107.037	1523.5	1018.2	680.85
$f_2 / (\text{t/h})$	0.0050	0.2128	0.1860	0.0306

A single goal optimization for each target can obtain the optimal value of each target. However, if both targets are optimized at the same time, it is impossible for both targets to reach the minimum value at the same time, but a trade-off solution, which is the significance of the multi-objective optimization algorithm. Table VI compares the optimal values for single-objective optimization of economic parameters with the optimal values for multi-objective optimization. The corresponding emission parameter is an emission parameter calculated using the value of the decision variable of the optimized economic parameter under a specific working condition. Taking full-speed working conditions as an example, if only economic indicators are considered, the optimal value after single-objective optimization is 1513.6 US \$/h, and the corresponding emission index is 0.2157 t/h. Hence, the fuel consumption is low at that time. The optimal value of the economic indicators optimized and determined by the ship's power distribution algorithm is 1523.5 US \$/h, and the optimal emission index is 0.2128 t/h. It can be seen that the fuel consumption and emission indexes cannot be reduced at the same time, but the emission indexes can be significantly reduced at the expense of a small part of the increase of fuel consumption. The economic index increases 0.65% slightly, while the emission index decreases 1.34%. Therefore, this shows that the multi-objective optimization model and the ship dynamics allocation algorithm can reduce the pollution emissions while taking into account the economic benefits, which fully reflects the effectiveness of the algorithm.

TABLE. VI OPTIMAL INDEXES AFTER SINGLE OBJECTIVE OPTIMIZATION FOR ECONOMY

Single Objective function	HB	FS	T	DP
$f_1/(\text{US \$}/\text{h})$	110.35	1513.6	1020.3	404.4
$f_2/(\text{t}/\text{h})$	0.0047	0.2157	0.1839	0.0857

VII. CONCLUSION

This paper designs a ship power distribution algorithm to solve the power distribution problem of hybrid propulsion ships. The innovation is that the method for dealing with the mixed integer constraints of ship crew coordination is given. The algorithm first performs the branch operation, decomposes the whole optimization problem into the optimization sub-problem, then calculates the effective scheme for the pruning operation, narrows the scope of the search domain, and uses the non-dominated sorting genetic algorithm to optimize the economic and emission indicators to obtain the Pareto frontier. All Pareto frontiers judge the dominance relationship, delete the governed solution and the repeated solution to get the Pareto frontier, and then use the Fuzzy decision method to obtain the final optimal solution. The optimization results show that the multi-objective optimization model and the ship dynamics allocation algorithm can reduce pollutant emissions while taking into account economic benefits.

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